Deriving long-run total costs

Production function: $q = K.L$

TC, AC, MC = ?

w, v = constants.

TC = min. cost of producing an output.

$TC = TC(q)$

optimum L, K.

$\frac{MP_L}{w} = \frac{MP_K}{v}$

$MP_L = \frac{\partial q}{\partial L} = \frac{\partial}{\partial L}[K \cdot L] = K$

$MP_K = \frac{\partial q}{\partial K} = \frac{\partial}{\partial K}[K \cdot L] = L$

$\frac{K}{w} = \frac{L}{v}$

$\boxed{\frac{Kv}{w} = L}$ — (1)

$q = K \cdot L = K \cdot \frac{Kv}{w} = K^2 \frac{v}{w}$

$\Rightarrow K^2 = \frac{qw}{v}$

$K = \left[\frac{qw}{v}\right]^{1/2}$ — (2)

$L = K\frac{v}{w} = \left[\frac{qv}{w}\right]^{1/2}$ — (3)

Deriving long-run total costs

Production function: $q = K.L$

$$TC = wL + vK = w \cdot \overbrace{\left[\frac{qv}{w}\right]^{1/2}}^{3} + v \cdot \overbrace{\left[\frac{qw}{v}\right]^{1/2}}^{2}$$

$$= (qvw)^{1/2} + (qvw)^{1/2} = 2\left[qvw\right]^{1/2}$$

$$AC = \frac{TC}{q} = \frac{2\left[qvw\right]^{1/2}}{q} = 2 \cdot \left[\frac{vw}{q}\right]^{1/2}$$

$$MC = \frac{dTC}{dq} = \frac{d}{dq}\left[2(qvw)^{1/2}\right] = \cancel{2} \cdot \frac{1}{\cancel{2}} \cdot \left[\frac{vw}{q}\right]^{1/2}$$

$$= \left[\frac{vw}{q}\right]^{1/2}$$

Example: Short-run costs

Try it yourself!

A firm producing hockey sticks has the following production function: $q = 2\sqrt{KL}$. In the short run, the firm's capital is fixed at K = 100. The rental rate of K is $v = \$1$ and the wage rate for L is $w = \$4$. → w

v

a. Calculate the firm's short-run total cost and average cost functions.

b. The firm's short run marginal cost function is given by SMC = q/50. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

c. Graph the SAC and the SMC curves for the firm. Indicate the points in part (b).

d. Where does the SMC curve intersect the SAC curve? Explain why the SMC curve will always intersect the SAC at its lowest point.

Example

a. Calculate the firm's short-run total cost and average cost functions.

Part a.

We know that $q = 2\sqrt{KL}$ and that K=100. Therefore,
$q = 2\sqrt{100L} \implies q = 20\sqrt{L} \implies \sqrt{L} = \frac{q}{20} \implies L = \frac{q^2}{400}$.

Therefore, $STC = vK + wL = 100 + \frac{q^2}{100}$, and

$SAC = \frac{STC}{q} = \frac{100}{q} + \frac{q}{100}$

Example

b. The firm's short run marginal cost function is given by SMC $= q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

Example

b. The firm's short run marginal cost function is given by SMC $= q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

Part b.

We know that SMC $= q/50$

If $q = 25$, then STC $= 106.25$,

Example

b. The firm's short run marginal cost function is given by SMC $= q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

Part b.

We know that SMC $= q/50$

If $q = 25$, then STC $= 106.25$, SAC $= 4.25$

Example

b. The firm's short run marginal cost function is given by SMC = $q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

Part b.

We know that SMC = $q/50$

If $q = 25$, then STC = 106.25, SAC = 4.25 and SMC = 0.5

If $q = 50$, then STC = 125, SAC = 2.5 and SMC = 1

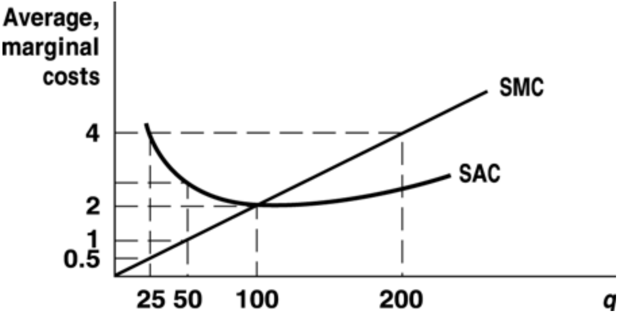
If $q = 100$, then STC = 200, SAC = 2 and SMC = 2

If $q = 200$, then STC = 500, SAC = 2.5 and SMC = 4

Example

c. Graph the SAC and the SMC curves for the firm. Indicate the points in part (b).

Part c.



Example

d. Where does the SMC curve intersect the SAC curve? Explain why the SMC curve will always intersect the SAC at its lowest point.

Part d.

SMC and SAC intersect at q = 100. As long as the marginal cost of producing one more unit is below the average cost curve, average costs will be falling. Similarly, if the marginal cost of producing one more unit is higher than the average cost, then average costs will be rising. Therefore, the SMC curve must intersect the SAC curve at its lowest point.